Rank Instability and Representation Repair for Automatic Modulation Classification under Structured Interference

Anonymous Author(s)

***Abstract*—Learned wireless receivers trade information-rich signal representations for compact front ends under tight inference budgets, yet the preferred representation can change with channel, mobility, and interference regime, so a single campaign leaderboard may not rank models stably. In source-disjoint simulations, the sealed family resolves an A5–A0 whole-configuration gain of 4.57 percentage points ($[3.65, 5.49]$ percentage points), while the isolated teacher and route-count interventions remain unresolved. Marginal scores favor the strong I/Q references, but a post-hoc severe-condition analysis inside the frozen campaign reverses the compact-versus-I/Q ordering by 9.74 and 10.46 percentage points. An independently generated severe-held evaluation then shows that this favorable ordering does not transport: under a seed-wise convention for paired inference, A5 has contrasts of -2.58 and -13.03 percentage points against IQFormer-inspired and MCLDNN, and the frozen overlap–SIR diagnostic surface underperforms SIR-only or constant predictors. On the same held regimes, a lightweight received-I/Q sidecar raises macro-F1 from 20.72% to 25.09%, a paired repair of 4.37 percentage points ($[2.96, 5.90]$ percentage points) in all ten seeds and all eight SNR strata—a consistent relative repair at an extreme operating point where every evaluated model remains low in absolute macro-F1. Prospectively frozen inference controls show that tested spectral width scaling does not reproduce the repair and that destroying received-I/Q information removes a large fraction of it, while removing the added fusion head removes the gain entirely; the repair is therefore best interpreted as a joint representation-and-fusion intervention rather than an isolated I/Q-information effect. The results frame AMC model comparison as a learned-receiver robustness problem: representation ranking can be regime-dependent, campaign-derived condition surfaces may not transport, and a lightweight information-rich path can recover part of the exposed degradation at low additional compute. These conclusions remain simulation-domain evidence rather than an over-the-air deployment claim.**

***Index Terms*—Automatic modulation classification, structured interference, rank instability, representation repair, domain shift, compact neural networks, interference overlap, vehicular Doppler.**

Evidence status. The original sealed family resolves one positive whole-configuration contrast (A5 versus A0), while the isolated teacher-form and route-count contrasts and the clean-retention gate do not pass. Campaign-level SNR–SIR, envelope, fusion, and capacity analyses are exploratory or post-hoc, whereas the probe, coverage, route-gating, and original sidecar experiments are prospective diagnosis or intervention evidence outside the sealed family. A prospectively frozen independent severe-held transfer plan characterizes the cross-regime transport boundary of the campaign-internal severe reversal; after that result, a second frozen plan specifies the sidecar-repair test before any sidecar inference on the same held set, and a third frozen plan specifies the attribution controls before any control inference. None of these post-campaign stages alters the original sealed family.

I. Introduction

Automatic modulation classification supports spectrum monitoring and blind receiver configuration by inferring a waveform's modulation from received samples. Deep AMC has progressed from convolutional I/Q classifiers [1], [2] to complex-valued, multi-stream, recurrent, and Transformer-like representations [3]–[7]. Compact architectures remain attractive when inference cost matters [8], [9], but average accuracy alone does not say where a compact representation is preferable.

This omission is consequential in mobile and vehicular receivers. Structured interference and mobility can change which representation is preferred under a fixed compute budget. A marginal score then hides both the rank changes and the risk that a campaign-internal favorable region will fail under a new regime. The issue extends beyond AMC model comparison. Learned wireless receivers often trade information-rich signal representations for compact front ends under tight inference budgets, while their deployment environments span channel, mobility, and interference regimes not represented by one training campaign. A static leaderboard is therefore insufficient whenever the preferred representation changes with operating condition. The more relevant design questions are whether such ranking relationships transport and whether lost robustness can be recovered without reverting to a full high-cost receiver.

This paper asks whether compact spectral and high-capacity I/Q rankings remain stable across structured-interference conditions. It then tests whether an apparent campaign-internal boundary survives independent severe-domain validation. When that interpretation fails, we ask whether the degradation can be diagnosed and repaired at the representation level. We answer using a frozen source-disjoint simulation campaign, artifact-only analysis, and prospective interventions. The base model, VIMD-Net, maps a complex STFT to three allocation routes and uses a training-only simulation-component teacher. No sealed contrast confirms either element individually. The contribution is the rank-instability evidence, its independent cross-regime transport test, and the representation-repair chain.

The contributions are:

- a controlled sealed A5–A0 whole-configuration contrast of 4.57 percentage points ($[3.65, 5.49]$ percentage points), while the isolated teacher-form and route-count effects remain unresolved; the result is not interpreted as a component-level or capacity-independent effect;
- a full SNR–SIR study showing that marginal model ordering can reverse inside the frozen campaign, motivating explicit ranking-stability analysis rather than a single global leaderboard;
- a source-disjoint independently generated severe-domain evaluation showing that the campaign-internal favorable ranking and its low-dimensional condition surface do not

transport uniformly across unseen jammer, mobility, and channel combinations; and

- a lightweight received-I/Q sidecar that produces a consistent 4.37-pp repair ($[2.96, 5.90]$ percentage points) on the same independent severe-held regimes at about 3% additional MACs, where targeted frozen-inference controls show that the repair is not reproduced by tested spectral width scaling and depends on received-I/Q information, but is best interpreted as a joint representation-and-fusion intervention rather than an isolated I/Q-information effect.

The claim is deliberately bounded. Compact spectral and high-capacity I/Q representations exhibit condition-dependent rank instability inside the frozen campaign, but the apparent severe boundary does not transport uniformly. The strongest surviving intervention evidence is that a lightweight received-I/Q path consistently repairs part of the independent severe degradation at an extreme operating point, and that the repair behaves as a joint representation-and-fusion intervention under frozen-inference controls.

II. Related Work and Claim Boundary

A. Strong and Efficient AMC Representations

CNN and recurrent AMC systems learn directly from I/Q sequences [1], [2], [5]; later work preserves complex structure, fuses temporal and spatial streams, or applies attention [3], [4], [6], [7]. Vehicular and mobile receivers have motivated reparameterized causal convolutions, cross-scale kernel selection with IQ-imbalance and carrier-offset compensation, and channel-robust cumulant features [10]–[12]. Compact and efficient AMC has received sustained attention for cost-sensitive deployment [8], [9]. These lines of work motivate strong I/Q-domain comparators and prevent us from presenting a lightweight architecture or a fusion scheme as the primary novelty.

B. Robust, Cross-Domain, and Interference-Aware AMC

Channel adaptation, masked or contrastive learning, and interference-tolerant features address different nuisance sources [13]–[16], while overlapped-signal classification, denoising masked autoencoding, and cross-domain prototypical adaptation populate the 2024–2026 frontier [17]–[19]. Interference-tolerant recognition [16] and tensor-based malicious-interference recognition for MIMO [20] address structured interference directly. Our setting differs: we neither reconstruct sources nor seek universal interference robustness; we study how the *ranking* of compact spectral versus high-capacity I/Q models changes with the interference condition itself.

C. Distinction: Rank Instability, Transport Boundary, and Repair

Knowledge-guided and multimodal preprints continue these directions [21]–[24]. Ranking changes under distribution shift are a known broader phenomenon: natural shifts can reorder apparent robustness [25], accuracy trends may change across shifted test sets [26], and benchmark suites expose heterogeneous distribution-shift failures [27]. Our contribution is therefore neither the generic possibility of ranking changes under distribution shift, nor another average-dominance claim, nor a universally valid selection surface. The novelty is the controlled wireless-receiver evidence chain that links campaign-level rank instability, an independently generated severe-domain transport test, and a low-cost joint representation-and-fusion repair evaluated on that same held domain. We study ranking instability, use a low-dimensional envelope as a diagnostic hypothesis, test its transfer under independent severe shift, and intervene on the exposed degradation. This is a positioning claim, not an exhaustive systematic review.

III. System Model and VIMD Configuration

A. Mobile Received Signal and Paired Views

We model a receiver-side AMC front end under terrestrial high-mobility conditions. For source sequence $s[n]$ and jammer $u[n]$, one received window is

$$x[n] = \mathcal{R}(\mathcal{H}_s(s)[n] + \kappa\mathcal{H}_u(u)[n] + w[n]), \tag{1}$$

where $\mathcal{H}_s$ and $\mathcal{H}_u$ are independently drawn fading and mobility operators (parameterized by carrier frequency, terminal speed, and 3GPP TR 38.901 [28] TDL-A/C/D profiles during training, with TDL-B/E held out for channel evaluation), κ scales the jammer contribution so that the realized SIR, defined as $10\log_{10}\big(E[|\mathcal{H}_s(s)|^2]/E[|\kappa\mathcal{H}_u(u)|^2]\big)$ on the simulator's tracked powers, matches the target value, w is receiver noise, and $\mathcal{R}$ applies the frozen receiver chain. The carrier frequency is $f_c = 5.9$ GHz, representative of intelligent-transport-system (ITS) and vehicle-to-everything (V2X) operation [29], and the complex baseband sampling rate is 1 MHz. Terminal speeds seen during training are 0, 60, 120, and 150 km/h, giving a maximum nominal Doppler shift of about 820 Hz; the held-speed split uses 180 and 250 km/h, i.e. up to about 1367 Hz, so speed and TDL profile are held out along separate axes. The held-speed split probes deeper intra-window non-stationarity: $T_c \approx 0.423/f_D$ falls to about 0.31 ms at 250 km/h, below the 1024 μs decision window.

Training pairs two independently impaired observations of the same immutable source; inference uses one view, so pairing is a training and statistical-alignment device, not additional test-time information.

Unanchored in-taxonomy cochannel collisions are excluded: without a physical target anchor, swapping two admissible emitters can preserve the mixture while changing the requested label.

B. Three-Route Spectral Allocation

Let $Z = \mathrm{STFT}(x) \in \mathbb{C}^{F\times T}$. The front end receives real, imaginary, and log-magnitude planes and produces logits A_s, A_j, A_o on the same lattice. The allocation masks are

$$\begin{aligned}&[M_s, M_j, M_o] = \mathrm{softmax}([A_s, A_j, A_o]),\\ &M_s + M_j + M_o = 1.\end{aligned} \tag{2}$$

An environment encoder produces a scalar ambiguous-route gate λ and a bounded bypass ρ. The modulation and jammer routes are

$$Z_m = (M_s + \lambda M_o + \rho) \odot Z, \tag{3}$$
$$Z_j = (M_j + (1-\lambda) M_o) \odot Z. \tag{4}$$

Because these weights are real and nonnegative, every retained coefficient keeps the mixture STFT phase. The bounded bypass ρ is a per-window scalar in the frozen range $[0.05, 0.35]$. It is added only to the modulation route; therefore the retained route weights remain nonnegative but sum to $1 + \rho$ rather than one, and the allocation is no longer an exact partition of the mixture power, i.e. $Z_m + Z_j \neq Z$ in general. This is the ratio-masking construction familiar from supervised source separation [30]–[33]; the elementary phase-preservation property does not imply source identifiability.

C. Simulation-Component Teacher

During training only, independently tracked post-channel target, jammer, and residual components define powers P_s, P_j, P_u on the student's lattice. With $q_k = P_k/(P_s + P_j + P_u)$, the fixed teacher is

$$M_s^\star = [q_s - q_j]_+, \qquad M_j^\star = [q_j - q_s]_+, \tag{5}$$
$$M_o^\star = q_u + 2\min(q_s, q_j). \tag{6}$$

The routes are nonnegative and sum to one; negligible-energy cells are assigned to the uncertainty route by the finite-precision implementation. We call this construction *physics-informed* because the supervision uses simulator-tracked components; it does not imply that the margin teacher or the three-route allocation is individually confirmed (Section V-A). The teacher is not needed at inference.

D. Training Objective

The total loss is

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\text{mod}} + \alpha\mathcal{L}_{\text{teacher}} + \beta\mathcal{L}_{\text{jammer}} + \gamma\mathcal{L}_{\text{link}} \\ + \eta\mathcal{L}_{\text{contrast}} + \theta\mathcal{L}_{\text{orth}},\end{aligned} \tag{7}$$

where $\mathcal{L}_{\text{mod}}$ is the modulation classification loss, $\mathcal{L}_{\text{teacher}}$ is a Jensen–Shannon [34] alignment between student and teacher masks, $\mathcal{L}_{\text{jammer}}$ and $\mathcal{L}_{\text{link}}$ constrain the jammer-family and link-quality auxiliary objectives, $\mathcal{L}_{\text{contrast}}$ is a supervised-contrastive loss [35] over exact-source positives that encourages condition invariance, $\mathcal{L}_{\text{orth}}$ is a route-orthogonality regularizer, and $\alpha, \beta, \gamma, \eta, \theta$ are nonnegative weights. The exact-source contrastive loss is an auxiliary training objective, not a headline contribution.

E. Information Contract and Complexity

At inference every model receives only the mixture; simulator-tracked target, jammer, and SIR quantities are training-only or post-hoc audit variables. VIMD-Net uses 39500 parameters and 41.8 million reported MACs.

F. Implementation Details

Unless otherwise stated, all models share the same source-disjoint cache, data budget, optimization budget, stopping rule, and validation-only selection criterion; model-specific components follow each variant's predefined configuration.

Input and front end. One example is a complex baseband window of 1024 samples (1024 μs at 1 MHz, 4 samples per symbol, root-raised-cosine roll-off 0.35), power-normalized by the frozen receiver automatic gain control. The spectral front end applies a two-sided STFT with $N_{\text{FFT}} = 64$, hop 16, a periodic Hann window of length 64, and no centering or zero padding, giving a 64×61 time–frequency lattice at 15.6 kHz per bin. The student and the fixed teacher share this lattice exactly.

Architecture width. The sealed configuration uses 24 spectral channels, embedding dimension 48, environment dimension 32, and dropout 0.05, for 39500 parameters. The prospective capacity ladder widens only these three numbers: the M tier uses (40, 80, 48) and the L tier (64, 128, 64).

Optimization and model selection. All fits use AdamW with an initial learning rate of 3×10^{-4}, weight decay 10^{-2}, batch size 16, mixed precision, gradient-norm clipping at 5.0, and a shared 30-epoch budget. Auxiliary losses follow a fixed staged schedule, and the learning rate decays on a cosine to a floor of 0.05 times its initial value once the staged schedule is complete. The reported checkpoint is chosen solely by validation modulation cross-entropy under a common eligibility window and 8-epoch early-stopping rule. Optimizer settings, schedule, and selection rule are identical for every model; no comparator received an architecture-specific sweep, and validation data are never used for evaluation.

Loss weights. In (7), $\alpha = 0.50$, $\beta = 0.25$, $\gamma = 0.05$, $\eta = 0.10$, and $\theta = 0.01$, with label smoothing 0.05 and contrastive temperature 0.12. The route-orthogonality regularizer is part of the A5 bundle but not isolated.

Seeds. The sealed campaign uses 10 algorithm seeds shared across all models, with seed-aligned initialization; matched-subset displays use the first five. Exact schedules, seed identifiers, teacher construction, and cache provenance are recorded in the reproduction package. Fig. 1 is a method illustration, not performance evidence.

IV. Evidence Protocol

A. Frozen Source-Disjoint Campaign

The artifact-locked campaign contains 12 model variants, 10 algorithm seeds, 120 fits, 11 evaluation splits, and 100000 training source sequences. Target classes are BPSK, PI/2-BPSK, QPSK, 8PSK, 16QAM, 64QAM, 256QAM, GMSK, CPFSK, and 4FSK, covering PSK, QAM, continuous-phase, and FSK families. Structured jammer families span tone, multitone, chirp, sweep, partial-band, pulse, comb, and out-of-taxonomy (cochannel and OFDM-like) modulated sources. TDL-A/C/D profiles are seen during training, while TDL-B/E profiles support held-channel evaluation. The evaluation grid spans eight SNR levels from -10 to 18 dB. The in-distribution split uses SIR levels -10 to 10 dB in 5 dB steps, whereas the hard-interference split spans SIR $\in \{-15, -10, -5, 0\}$ dB, so

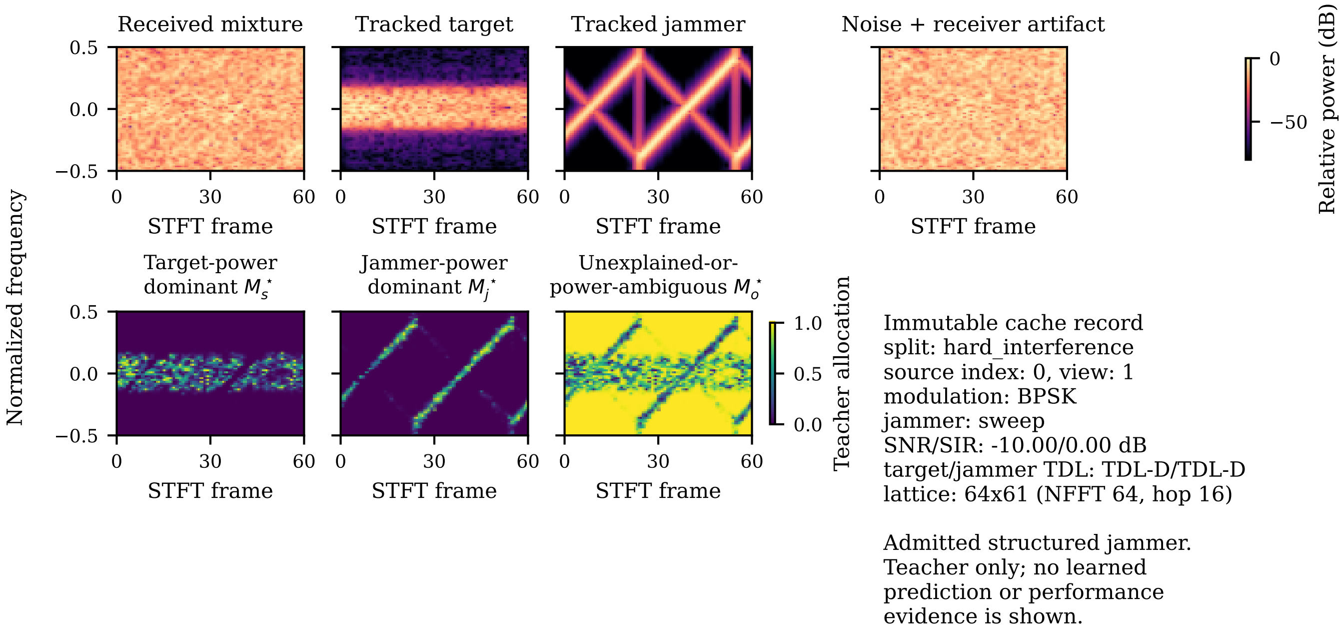

Fig. 1. A deterministic cache record and the fixed simulation-component teacher of Eqs. (5)–(6). Top row: received mixture, tracked target, tracked jammer, and noise-plus-artifact spectrograms; bottom row: the target-dominant, jammer-dominant, and uncertainty route masks. The teacher is a training-only construction from simulator-tracked components; it is unavailable at inference and is not performance evidence.

SIR $= -15$ dB occurs only in the hard-interference split and SIR $= +5$ and $+10$ dB only in the in-distribution split. Each hard-interference SNR–SIR evaluation cell of Fig. 2 contains 131–181 source-disjoint test windows (mean 156; all ten classes, 5–26 per class), read from the frozen prediction cache. The envelope aggregates of Section V-C use a coarser partition of the same predictions (216–279 source-disjoint windows per held cell, mean 250). The fit-domain division (in-distribution plus hard-interference cells) is fixed in the exploratory analysis protocol before any envelope is fitted. Jammer family, speed, channel, combined OOD, receiver stress, and clean-retention splits share source identities across model comparisons. The receiver-stress split (ADC quantization and per-emitter synchronization) is retained in the artifact package but does not enter any sealed, envelope, or reported selector claim.

B. Comparator Fairness

Table I summarizes the comparators. All models are retrained under the same source-disjoint protocol; "IQFormer-inspired" and "MCLDNN" denote architecture-faithful unified-retraining references, not exact reproductions of every original training recipe. Comparator fairness is enforced at the budget rather than at the tuning level: every model sees the same training source count, the same epoch budget, the same early-stopping patience and checkpoint criterion, and the same validation protocol. CSSL is retained as an adaptation-oriented reference rather than a tuned upper bound; its 8.63 million parameters but only 155.3 million MACs reflect aggressive encoder downsampling and a large projection head, so parameter count alone is a poor cost proxy for this comparator. Figures label this comparator "IQFormer-insp."; "A5" and "VIMD-Net" denote the same sealed configuration,

TABLE I
COMPARATOR AND COMPLEXITY SUMMARY (HARD-INTERFERENCE SETTING).

Model	Domain	Params	MACs	Seeds
A0	spectral	8458	6.3 M	10
A5 / VIMD-Net	spectral allocation	39500	41.8 M	10
sidecar	spectral + received I/Q	46794	43.1 M	10
MCLDNN	I/Q	406070	398.2 M	10
IQFormer-inspired	multimodal I/Q	354984	355.6 M	10
CSSL	I/Q (adaptation)	8631948	155.3 M	10

and "CSSL" abbreviates the contrastive self-supervised learning reference of [15], whose released code snapshot [36] we retrain under our protocol. Because this campaign retrains CSSL under the common supervised budget without executing the released self-supervised pretraining stage, it is reported as a supervised adaptation reference rather than a reproduction of the full released recipe.

As a public-benchmark comparator-fidelity check, the MCLDNN and IQFormer implementations were additionally evaluated on the public RML2016.10a benchmark [1] under the protocol of the official benchmark result table of the IQFormer release [37]: the 11-class set, the full -20 to 18 dB SNR grid, per-stratum $600/200/200$ splits seeded at 233, no augmentation or input-length change. Over three algorithm seeds, all-SNR test accuracy is 58.55% for MCLDNN (seed range 58.1–59.4%) and 65.23% for IQFormer-inspired (64.8–65.8%; seed ranges, not confidence intervals); the reference-release values, averaged over the same twenty SNR bins, are 62.05% and 64.19%. IQFormer-inspired therefore closely matches the reference-release all-SNR accuracy ($+1.04$ pp), whereas MCLDNN remains moderately lower (-3.50 pp) while staying in the same broad all-SNR range. Per-SNR deviations of up to about 8 pp remain, so E1 is range-level comparator-implementation fidelity, not pointwise reproduction nor evidence for the simulator-domain envelope.

C. Original Sealed Confirmatory Family

The sealed family contains three hard-interference contrasts: the complete configuration versus A0, the margin teacher versus a proportional-power teacher, and the tri-route model versus a dual-route control. A5 differs from A0 in allocation, teacher, auxiliary objectives, residual bypass, and capacity (about $4.7\times$ parameters), so A5–A0 is a whole-configuration effect.

D. Exploratory and Tier-2 Evidence Classes

All SIR-cell, overlap, fusion, condition-transfer, probe, repair, and capacity analyses are exploratory. In particular, the severe SIR stratification is post hoc. Tier-2 experiments were designed after inspecting the frozen evidence and are outside the frozen confirmatory family; they cannot retroactively change a sealed conclusion. The clean-retention gate was preregistered separately and did not pass; clean behavior is therefore treated as a boundary to explain and repair, not as a preserved property of the sealed model. The independent severe-domain transfer plan was frozen before any inference on the new held set. After that evaluation revealed limited cross-regime transport, a second analysis plan froze the sidecar-repair hypothesis, primary contrast, and decision rule before any sidecar inference on the same held data. A third plan then froze the inference-only attribution controls of Section V-F before they were executed. Local plans, hashes, and execution records establish this ordering; no external timestamping service was used. All three remain post-campaign diagnostic evidence and do not alter the sealed family.

E. Statistical Procedures

The sealed family uses a joint non-studentized max-absolute-deviation hierarchical paired bootstrap with 10000 draws, resampling class-stratified source clusters while keeping algorithm seeds as fixed blocks, so pairing is preserved at the source level. The design resolution of 1.31 percentage points is the simultaneous minimum detectable effect at 80% power under this joint interval, an exploratory calibration rather than a sealed significance threshold. Because seeds are fixed blocks, the reported intervals are conditional on the realized seed set; a seed-randomization variant resampling the ten realized seeds with replacement keeps the A5–A0 contrast positive (95% percentile interval $[4.02, 5.19]$ percentage points) while the teacher-form and route-count contrasts span zero ($[-0.15, 0.42]$ and $[-0.15, 1.04]$ percentage points). In the campaign's severe row, all 10 seeds share the same sign; the sign-flip permutation probability 0.0018 is a permutation test over the source-paired samples. The SIR $= -15$ dB aggregate is computed from source-paired predictions pooled across the eight SNR conditions before macro-F1 aggregation; it is therefore not the arithmetic mean of the eight displayed cell-level differences in Fig. 2. The severe-held levels and repair contrasts of Section V-F instead average per-seed macro-F1 values with equally weighted regimes (the seed-wise paired convention); the two conventions are never mixed within a single reported contrast.

TABLE II
SEALED CONFIRMATORY HARD-INTERFERENCE FAMILY. DIFFERENCES ARE CANDIDATE MINUS REFERENCE IN MACRO-F1 PERCENTAGE POINTS.

Contrast	Diff.	Sim. low	Sim. high
Full configuration vs A0	4.57	3.65	5.49
Margin vs proportional teacher	0.16	-0.76	1.08
Tri-route vs dual-route	0.44	-0.47	1.36

V. RESULTS

A. Positive Sealed Whole-Configuration Contrast and Bounded Component Effects

Table II reports the complete sealed family. The complete configuration improves over its shared A0 backbone by 4.57 percentage points, with a simultaneous interval of $[3.65, 5.49]$ percentage points, about 3.5 times the design resolution of 1.31 percentage points. The teacher-form and route-count interventions do not clear zero individually; under the same joint interval their simultaneous upper bounds are 1.08 and 1.36 percentage points, respectively. Thus the configuration-level effect is resolved but cannot be assigned to either isolated component at this resolution, and because the family rule required all three contrasts, the family gate did not pass. Because the A5–A0 interval is part of the same simultaneous family-wise band, its positive whole-configuration contrast remains a valid family-wise-controlled result even though the prespecified joint family gate does not pass. At the same parameter tier, the residual-free A7 variant is nominally above A5 by 0.21 percentage points in the marginal hard-interference aggregate (Fig. 3); treating the ten realized seeds as a random effect gives a 95% percentile interval of $[-0.19, 0.62]$ percentage points, so the difference is not resolved at this design's power; we therefore attribute the compact frontier position to the VIMD family at this budget, not uniquely to any single component.

B. Campaign-Internal Rank Instability across the SNR–SIR Plane

Marginal hard-interference macro-F1 is 0.4406 for A5, 0.4599 for MCLDNN, and 0.4872 for IQFormer-inspired. Those averages favor the larger models. Figure 2 shows why no single campaign-level ranking is adequate: the sign changes with SNR and SIR, including a severe-row reversal. Section V-C summarizes this campaign structure without assuming that it transfers.

At SIR $= -15$ dB, in a post-hoc analysis, A5 gains 9.74 percentage points over MCLDNN ($[7.52, 12.14]$) and 10.46 percentage points over IQFormer-inspired ($[6.78, 14.29]$); these two headline intervals are marginal stratified-bootstrap intervals and are not multiplicity-adjusted. All 10 seeds have the

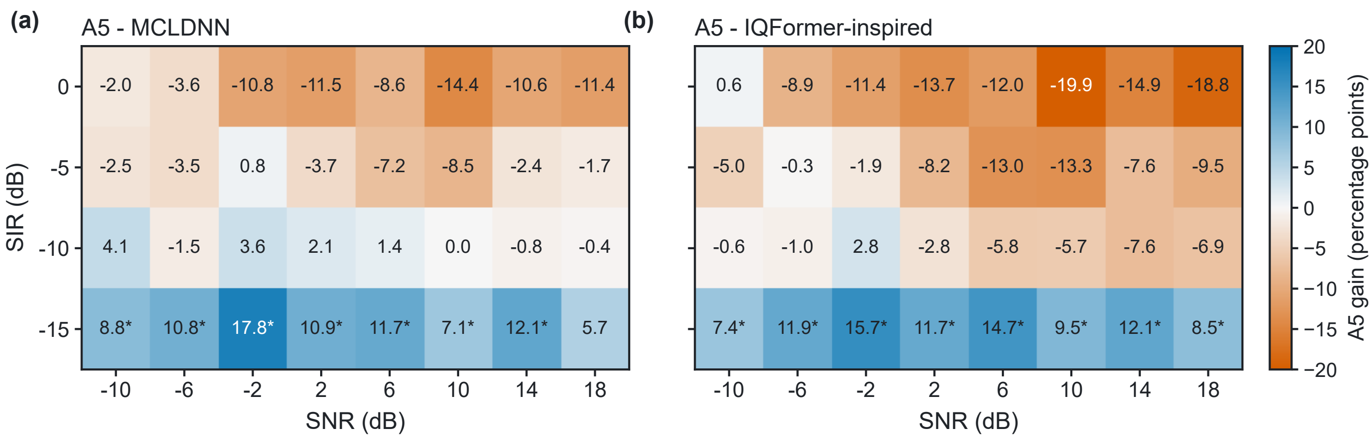

Fig. 2. Campaign-internal post-hoc rank instability over all hard- interference SNR–SIR cells. Each entry is A5 minus the named I/Q reference in macro-F1 percentage points. Blue cells favor A5 and orange cells favor the reference; color and printed values encode the same sign. A star marks a positive unadjusted paired lower interval; Holm-surviving cells are reported in the text. The display does not imply transport outside the frozen campaign.

same sign, and the sign-flip permutation probability is 0.0018. Of 64 inspected cells, 12 survive Holm correction; all are on the severe SIR row. These checks stabilize the reversal inside the campaign without converting the post-hoc analysis into confirmation. Section V-F prospectively tests a broad transport interpretation and rejects it on independently generated severe-held data.

C. Campaign-Diagnostic Overlap–SIR Envelope

The campaign-internal rank instability motivates a low-dimensional diagnostic surface. Rather than assuming a universal ranking, the surface summarizes how gain varies with target-support jammer overlap and SIR inside the frozen campaign. For reference b, we fit

$$\Delta_b = \beta_{0,b} + \beta_{1,b} o + \beta_{2,b}\,\mathrm{SIR}, \tag{8}$$

where Δ_b is A5 minus the reference macro-F1 and o is the *target-support jammer overlap*. We avoid the more common "spectral occupancy" because o is not an occupied-bin or occupied-bandwidth fraction. The covariate is defined from the simulator-tracked target and jammer components rather than from the mixture. For each window we compute the STFT power of the tracked target and of the tracked jammer separately, order the time–frequency cells by descending target power, and take the target's 99%-energy support $\mathcal{S}$ as the smallest set of cells whose cumulative target power reaches 99% of the total. The overlap is then the fraction of the jammer's total STFT power that falls inside $\mathcal{S}$:

$$o = \frac{\sum_{(f,t)\in\mathcal{S}} P_j(f,t)}{\sum_{(f,t)} P_j(f,t)}. \tag{9}$$

Thus $o = 0.8$ means that 80% of the jammer's energy lies within the target's dominant support, not that the jammer fills 80% of the plane. The cell-level covariate is the mean of o over the windows of that cell.

Equation (9) is evaluated on a fixed analysis lattice, a Hann-windowed 128-point STFT with hop 32, deliberately independent of the 64-point VIMD front-end lattice of Section III-F: the front-end lattice is what the model sees, whereas the analysis lattice is a measurement instrument for the physical audit. The overlap is computed once per window when the cache is built, before any model is trained; it is stored in the sealed cache, checksummed in the cache manifest, and read unchanged by every model, split, and comparison in this paper. It is never an input feature of VIMD-Net or of any comparator, and it is not tuned against any model's predictions.

The fit uses 32 aggregates, reconstructible from the protocol rather than asserted. A cell is one (split, SIR stratum, overlap quartile) triple that retains at least 150 windows and all ten classes. The `hard_interference` split contributes 4 SIR strata $(-15, -10, -5, 0\text{ dB}) \times 4$ overlap quartiles $= 16$ cells, and `id_test` contributes 16 further cells over its five SIR strata; its lowest-overlap quartile is dominated by jammer-free windows ($o = 0$, SIR $= 0$ dB cache sentinel) and clears the window floor only at 0 dB, which fixes the per-stratum counts. The primary held set is the 80 finite-SIR interfered cells formed the same way from unseen jammer, unseen speed, held-out channel, and combined OOD (4×5 SIR strata $\times$ 4 quartiles $= 80$). Clean retention is deliberately outside this primary envelope because a jammer-free window has no physical finite SIR. Overlap and SIR are nearly orthogonal across the fitting cells (Pearson $r = -0.003$; -0.03 over all 113 cells), so strong linear collinearity between the two fitted covariates is not evident. We nevertheless interpret Eq. (8) as descriptive rather than causal; low pooled correlation bounds collinearity, not confounding.

Against IQFormer-inspired, the fitted overlap coefficient is 14.38 percentage points per unit overlap and the SIR coefficient is -1.036 percentage points per dB. Magnitude skill is scored against the constant predictor that outputs the mean gain over the fitting cells, evaluated unchanged on the held-regime cells: RMSE 6.23 against 8.37 percentage points ($R^2_{\mathrm{skill}} = 0.45$), and 8.10 against 9.63 percentage points for MCLDNN ($R^2_{\mathrm{skill}} = 0.29$). A covariate ladder isolates each predictor: SIR-only and overlap-only envelopes attain held RMSE 6.84 and 7.46 percentage points against IQFormer-inspired (8.71 and 8.68 percentage points against MCLDNN), so the incremental overlap effect beyond SIR is not separately resolved by the paired bootstrap ($P(\Delta\text{MSE} > 0) = 0.94$ and 0.85, with both paired intervals including zero). Transport tests then bound the surface's role. Leaving out the -15-dB SIR stratum during fitting does not confirm an overlap increment: for MCLDNN the full surface reduces held-out RMSE relative to the constant baseline but predicts the correct sign in only two of four held-out cells, whereas the SIR-only surface preserves all four signs; for IQFormer-inspired the full surface is no better than SIR-only. The independently generated severe-held regimes of Section V-F likewise do not support the surface: envelope sign accuracy there falls below SIR-only and constant predictors. We therefore retain Eq. (8) only as a diagnostic summary of campaign-internal rank variation, not as a transportable switching rule.

Solving Eq. (8) for zero gain yields break-even overlap references that rise as interference becomes less severe; these are simulator-domain descriptive references conditional on an overlap estimate, not deployment-ready control thresholds. The preregistered mechanism expectation was non-increasing gain with overlap; the sealed directional test rejected that sign, and exploratory quintiles instead show monotonically increasing A5–CSSL gain. The positive overlap direction is not specific to this adaptation reference ($+19.16$ percentage points per unit overlap against MCLDNN); we return to it in Section VI-B.

D. Complexity and Complementarity

A5 remains useful despite unstable ranking because its errors differ from those of the I/Q models. Late probability fusion improves over the stronger constituent by 4.71 percentage points with IQFormer-inspired and 5.70 percentage points with MCLDNN. The corresponding same-model seed-ensemble controls improve by only 1.49 and 1.69 percentage points, so the cross-model fusion gains exceed the ensemble controls by 3.22 and 4.01 percentage points. This is supporting evidence of complementary decisions, not marginal superiority.

Table I gives the corresponding parameter and MAC budgets; compactness is stated as a parameter and MACs property rather than an end-to-end inference-cost claim.

E. Clean-Boundary Diagnosis

The cross-regime severe degradation makes representation robustness central to interpretation. The earlier preregistered clean-boundary diagnosis independently tests whether the compact spectral representation discards information that becomes important under distribution shift. The clean-retention split contains all ten modulations, but the two training views expose jammer-free windows for only 4 classes (PI/2-BPSK, 8PSK, 256QAM, and CPFSK); the remaining 6 classes (BPSK, QPSK, 16QAM, 64QAM, GMSK, and 4FSK) are a condition-transfer test. The four covered classes are marked without a star in Fig. 4, where a star denotes a class whose jammer-free condition was not seen during training. Transfer failure occurs in $27/27$ uncovered compact-spectral class–model cells (100%) versus $1/9$ I/Q-reference cells (11%). The clean-retention gate did not pass. Figure 4 localizes the failure rather than hiding it in an aggregate.

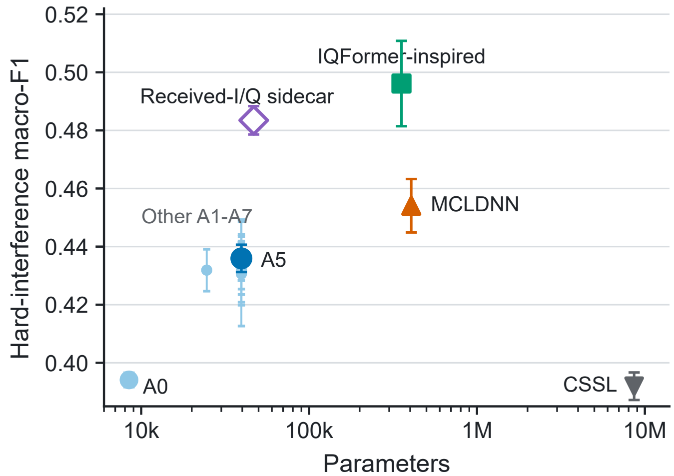

Fig. 3. Hard-interference macro-F1 versus parameter count on the matched five-seed subset. Error bars are ±1 seed standard deviation. Blue marks denote the spectral family, green IQFormer-inspired, orange MCLDNN, gray CSSL, and purple the received-I/Q sidecar. Filled marks are sealed models; the open sidecar diamond is exploratory and outside the sealed family. The comparison is descriptive and regime-specific.

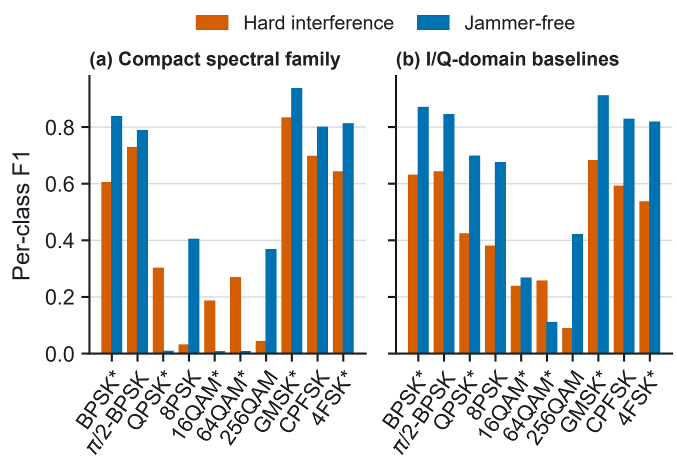

Fig. 4. Clean condition-transfer taxonomy. Bars show family-mean per-class F1 under hard interference (hatched orange) and jammer-free evaluation (blue) for the compact spectral family and I/Q-domain baselines. A star marks a class whose jammer-free condition was absent from training; it is not a significance marker. The same aggregation and scale are used in both panels.

We then followed a prospective decision chain. Frozen-checkpoint linear and MLP probes used a correctly reconstructed jammer-free counterfactual and source-disjoint clean cross-validation; probe performance is the unweighted mean F1 over QPSK, 16QAM, and 64QAM (chance ≈ 0.1 on ten classes). The preregistered decision rule declares a representation *information-present, head-limited* when this mean exceeds 0.25 and *representation-limited* otherwise, requiring *all* design–seed cells to exceed the threshold. For A5, five of the six design–seed cells (three algorithm seeds $\times$ the reconstructed jammer-free counterfactual and clean cross-validation probes) lie below the 0.25 threshold; the sole cell above it is borderline at 0.256, so we record the representation-limited side. The decision is unchanged at a threshold of 0.30 but flips at 0.20. The strong I/Q baselines exceed the threshold in

18 of 18 cells. The evidence therefore leans toward missing constellation information in the compact spectral representation rather than a classification head that fails to use available information.

Adding clean windows for every class tests the simpler coverage explanation. It changes clean macro-F1 by only 0.15 percentage points ($[-0.31, 0.63]$) and hard macro-F1 by -0.30 percentage points ($[-0.91, 0.27]$). Coverage alone is therefore insufficient. A preregistered negative control gates the modulation route using predicted interference presence. Relative to A5, it changes clean macro-F1 by -0.30 percentage points ($[-0.66, 0.08]$) and hard macro-F1 by -0.18 percentage points ($[-0.72, 0.35]$). The learned gate averages 99.994% on clean windows and 99.991% under hard interference, saturating near the modulated route; this arm is a negative control, not standalone evidence that a functional mask fallback was learned.

F. Representation Repair under Independent Severe Shift

Widening the spectral model without changing its representation domain first separates capacity from representation. M and L improve the campaign hard split, but the widest tier still trails IQFormer-inspired marginally and on clean retention. Tested spectral width scaling alone therefore does not remove the representation boundary.

A prospectively frozen, independently generated severe-held set then tests broad transport. At SIR $= -15$ dB, R1 combines held jammer families with 180-km/h mobility over training channels, while R2 combines held channels with the same jammers at 250 km/h. Each regime spans eight SNRs and 6400 source-disjoint examples (12800 total); every checkpoint remains frozen. The campaign-internal corner does not transfer under either aggregation convention. Envelope sign accuracy falls to 0.578 and 0.406, while SIR-only or constant predictors beat the full surface. This experiment therefore establishes a cross-regime transport boundary for both the corner and the -15 dB diagnostic surface.

The sidecar pools mean, standard deviation, and maximum statistics from a two-channel depthwise-separable received-I/Q branch, fuses its 24-dimensional embedding residually with the spectral path, and adds a side head. It uses only the received mixture and adds about 3% compute over A5.

On the same independent severe-held set, the frozen sidecar raises seed-mean pooled macro-F1 from 20.72% to 25.09%, a paired gain of 4.37 percentage points ($[2.96, 5.90]$). Repair is positive in all 10 seeds and all 8 SNR strata. The result is a consistent 4.37-pp relative repair at an extreme operating point where all evaluated models remain low in absolute macro-F1; it should therefore be interpreted as relative robustness repair rather than deployable classification accuracy at this operating point. Both severe-held validations remain exploratory post-campaign evidence outside the sealed family.

The prospectively frozen controls in Table III separate tested spectral width scaling from received-I/Q information and the added fusion head. The repair depends on both the received-I/Q path and its residual side head; these inference interventions diagnose information dependence but do not establish causality. Under the frozen decision rule the pattern is mixed, and the repair is best interpreted as a joint representation-and-fusion intervention rather than an isolated I/Q-information effect.

VI. Discussion

A. What the Rank Instability Establishes

The stable conclusion is not that A5 owns a severe-interference region. Model ranking is condition- and regime-dependent. Marginal scores favor the strong I/Q references, the frozen campaign contains a severe reversal, and an independent regime reverses that result again. This pattern supports bounded model comparison, not a transportable deployment selector.

B. Why the Diagnostic Envelope Does Not Generalize Uniformly

Equation (8) has skill over 80 held cells inside the frozen campaign, but that skill is heterogeneous by regime. The incremental overlap effect beyond SIR is not separately resolved, and leave--15-dB-out behavior depends on the reference. The preregistered directional hypothesis also fails: gain increases rather than decreases with overlap inside the campaign. The surface is therefore a diagnostic hypothesis, not a causal equation or a universal switching rule. Its overlap variable additionally requires simulator-tracked target and jammer components and is not directly observable at an uncooperative receiver.

C. Cross-Regime Transport Boundary

The independent severe-held experiment was fixed before inference and was allowed to reject the campaign interpretation: it covers unseen jammer, mobility, and channel combinations at the same severe SIR level, and its negative result is evidence about the boundary of the original finding, not a post-hoc exclusion. It also prevents the in-campaign RMSE and break-even references from being promoted to broad severe-domain transport claims. The boundary is stated without a deployment switching claim: the campaign indicates where the compact model was competitive, not where a receiver should switch.

D. Representation Repair

The intervention chain constrains, but does not prove, the explanation. Frozen probes point toward missing constellation information; coverage and route- gating controls do not support simpler exposure or fallback explanations as sufficient; the width ladder shows that tested spectral width scaling alone is insufficient inside the campaign and does not reproduce the severe-held repair. The severe-held inference controls further show that the repair depends on received-I/Q information but that the added fusion head is also necessary; the repair is therefore best interpreted as a joint representation-and-fusion intervention rather than as information preservation alone. The remaining gap to MCLDNN and the low absolute operating point limit the claim to relative repair, not uniform superiority or causal confirmation.

TABLE III

INDEPENDENT SEVERE-HELD TRANSPORT, REPAIR, AND FROZEN-INFERENCE CONTROLS. POSITIVE CONTRASTS FAVOR THE ROW MODEL; CONFIDENCE INTERVALS ARE SEED-WISE PAIRED 95% INTERVALS.

(a) Full ten-seed severe-held evaluation

Model or intervention	Macro-F1 (%)	Contrast or interpretation
A5 spectral baseline	20.72	Reference
IQFormer-inspired	23.29	$+2.58$ percentage points vs. A5
MCLDNN	33.75	$+13.03$ percentage points vs. A5
Received-I/Q sidecar	**25.09**	$+4.37$ percentage points $[2.96, 5.90]$
Sidecar, I/Q zeroed	12.80	-12.28 percentage points vs. full sidecar
Sidecar, I/Q shuffled	16.25	-8.84 percentage points vs. full sidecar
Sidecar, side head removed	7.09	Gain eliminated

(b) Matched five-seed capacity control

Model	A5	Width M	Width L	I/Q sidecar	Seeds
Macro-F1 (%)	20.72	21.77	20.89	**26.11**	5

E. Vehicular and Learned-Receiver Design Implications

At $f_c = 5.9$ GHz and the held-speed conditions, the channel can evolve materially within the 1024-μs decision window: the nominal coherence time at 180 and 250 km/h is about 42% and 30% of the window, increasing intra-window nonstationarity. Representation robustness is therefore a receiver-design issue, not only a classifier-ranking issue. More generally, the results caution against fixing a learned receiver architecture or switching rule solely from one campaign's marginal leaderboard or condition surface. When compact processing is desirable for compute reasons, preserving a small information-rich signal path may offer a lower-cost robustness mechanism than either committing to the compact representation alone or reverting to a full high-cost I/Q receiver. The sidecar illustrates this tradeoff within the present simulation domain; it is not an over-the-air deployment claim.

VII. LIMITATIONS AND REPRODUCIBILITY

The evidence has nine explicit limits. First, all results are from source- disjoint TR 38.901 TDL-profile simulations. They are not measured over-the-air or complete MAC-/network-layer V2X evidence. Second, the original sealed family resolves only the A5–A0 whole-configuration contrast, which includes a capacity difference; the isolated teacher and route-count contrasts and the clean-retention gate do not pass. Third, the independent severe-held validation shows that the campaign-internal corner and the frozen diagnostic surface do not transport broadly. Fourth, the sidecar repair and the three attribution controls remain prospectively frozen post-campaign diagnostic evidence outside the sealed family. Fifth, IQFormer-inspired and CSSL run to the shared 30-epoch limit in all ten seeds, so their comparisons retain a budget-sufficiency caveat; MCLDNN early-stops in every seed. Post-campaign M, L, and sidecar training histories are summarized in the Supplement; they all reach the shared epoch budget, although their selected epochs can precede the final epoch. Sixth, both strong I/Q comparators are unified-retraining references, and the public-benchmark check establishes range-level implementation fidelity rather than exact recipe reproduction. Seventh, all severe-held comparisons inherit the original campaign checkpoint-selection rule: checkpoints were selected on campaign validation loss rather than the independent held regimes, which preserves a clean frozen OOD evaluation but may interact differently with model-specific optimization and domain shift. Eighth, the teacher and overlap covariate require simulator-tracked components. The overlap is not directly available at an uncooperative receiver, and its break-even values are descriptive references. Ninth, the probe, exposure, route-gating, width, sidecar, and frozen-inference control chain is consistent with a joint representation-and-fusion repair but does not establish a causal mechanism.

Reproducibility is enforced at the artifact boundary: each manuscript macro stores the source path, row selector, evidence class, precision, and source SHA-256; every figure is rendered from a CSV with its own provenance manifest; and the resolved macros, figure-source CSVs, per-seed aggregates, configurations, and the full provenance manifest are released upon acceptance. The frozen run stores cache identity, checkpoints, source-aligned predictions, paired statistics, and execution records, preventing a favorable Tier-2 result from overwriting an unfavorable sealed gate.

VIII. CONCLUSION

Compact spectral and high-capacity I/Q AMC representations do not admit one stable ordering across structured vehicular interference regimes. The sealed campaign resolves a positive A5–A0 whole-configuration contrast but not the isolated teacher or route-count effects, and a post-hoc severe-condition reversal inside the campaign motivates explicit rank-stability analysis. An independently generated severe-domain evaluation then shows that the favorable ranking and the diagnostic overlap–SIR surface do not transport uniformly across unseen jammer, mobility, and channel combinations. On the same frozen held regimes, a lightweight received-I/Q sidecar improves A5 by 4.37 percentage points ($[2.96, 5.90]$ percentage points) at about 3% additional MACs, with consistent gains across all evaluated seeds and SNR strata, while every evaluated model remains low in absolute macro-F1 at this operating point. Together with the probe, exposure, route-gating, width, and severe-held inference controls, the evidence supports a joint representation-and-fusion interpretation of the repair without isolating an I/Q-information-only effect.

The practical lesson is therefore not a universal switching boundary, but the importance of evaluating ranking stability and preserving low-cost complementary signal information when compact learned receivers face operating-regime shift.

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